Phase 2 - Literature Review: Dental Clinic Management System

2.1 Introduction

This review examines published research, official documentation and existing systems to justify the design of a web-based Dental Clinic Management System (DCMS) built as an educational prototype. It covers five areas: patient and dental records, appointment scheduling, treatment documentation, health-data security and usability.

Electronic dental records (EDRs) are widespread at practice level, but they are often optimised for charting and procedure codes rather than for exchange with wider health systems [1]. EDRs are also not a simple transposition of paper records: they document medical and dental history together with detailed consultation information [2]. These two points shape the scope below. The prototype records and retrieves clinical information well, and does not attempt national-level integration.

All sources were retrieved on 23 October 2026. Statements that depend on a source carry a numbered citation, listed in Section 2.9.

2.2 Review Questions

ID	Question	Answered in
RQ1	What core functions do clinic and dentalpractice management systems provide?	2.5 (A, C), 2.6
RQ2	How do electronic patient records support continuity and retrieval of clinical information?	2.5 (A)
RQ3	How are appointments scheduled, and how are changes, conflicts and no-shows handled?	2.5 (B)
RQ4	Which security and privacy controls apply to systems holding sensitive health data?	2.5 (D)
RQ5	Which features fit the scope and resources of this student project?	2.5 (E), 2.7, 2.8

2.3 Search Strategy

All searches were run on 23 October 2026 through a general web search engine, then each hit was opened by title and abstract on its own site (PubMed Central, BMC, Wiley, MDPI, NIST, Open Dental and OpenEMR documentation). Up to ten ranked results per query were screened; exact counts were not recorded. **Limitation:** this was not a Google Scholar or PubMed query log. The team should repeat the queries below in those databases and record result counts to meet the reproducibility requirement of the template.

#	Query	Target
1	electronic dental records systematic review	RQ1, RQ2
2	appointment scheduling outpatient clinic no-show systematic review	RQ3
3	role-based access control healthcare information systems review	RQ4
4	Egypt Personal Data Protection Law No. 151 of 2020 health data	RQ4
5	NIST SP 800-53 Rev. 5 access control audit and accountability least privilege	RQ4
6	usability evaluation electronic dental record user satisfaction study	RQ5
7	dental treatment planning documentation electronic record proposed versus completed procedures	RQ1, RQ2

8	Open Dental manual appointments, treatment plan and chart modules	2.6
9	Open Dental user permissions security groups manual	2.6
10	OpenEMR documentation scheduling, access control (ACL), audit	2.6

2.4 Inclusion and Exclusion Criteria

The criteria in the Phase 2 template were applied as written: peer-reviewed research, official standards and official product documentation were included; unattributed pages and marketing claims were excluded. The decisions below show how that played out.

Source type	Decision	Reason
Peer-reviewed reviews and studies (BMC Oral Health, Health Policy, Health Science Reports, J Healthcare Eng, Entropy)	Included	Methods and findings are stated
Official documentation (opendental.com manual, OpenEMR wiki, NIST SP 800-53)	Included	Primary source for system features and controls
Software-review sites (GetApp, SelectHub, Techjockey)	Excluded for feature claims	Aggregated opinion and marketing, no stated method
Vendor-style blog posts on RBAC and OpenEMR	Excluded	No identifiable reviewer, written to sell services
Wikipedia page on Open Dental	Excluded	Tertiary; the official manual was used instead
JMIR preprint of the Ballester review	Excluded	Not peer-reviewed; the published BMC version was used
Law-firm summaries of Egyptian Law 151 of 2020	Included as secondary , flagged	Official gazette text should be checked before final submission

2.5 Thematic Review

Theme A — Patient and Dental Records (RQ1, RQ2)

Dental data is often siloed in a mix of paper and electronic records that do not interoperate with wider health systems, and most practice-level EDRs are built for odontograms, radiographs and procedure codes rather than reuse beyond clinical care [1]. Standardised coding of dental diagnoses developed late: in the early 2000s there was no consensus nomenclature for diagnoses and treatment outcomes, and a scoping review of 44 articles now summarises the work on clinical coding and EDR data capture
[2].

On content, a qualitative comparison of paper and computer-based dental records defined a baseline dental record. The UK Faculty of General Dental Practitioners guidance it cites requires demographics, medical and dental history, an examination of the dentofacial area, an initial dental examination, a treatment plan and progress notes [3]. The same study found that using paper and electronic records in parallel can cause inconsistencies in documentation [3].

On retrieval, a study with 20 dental students and 200 digitised records found that the electronic interface surfaced a highlighted alert about conditions needing attention before treatment, and every student noticed it on opening the record [4]. The sample is small, so this supports a design idea and is not a general effect.

Design implication: use the baseline record elements [3] as the minimum patientrecord fields, show a medical-alert banner on opening a record [4], and keep one electronic source of truth to avoid parallel-record inconsistencies [3].

Theme B — Appointment Scheduling (RQ3)

Non-attendance is the main operational problem. A systematic review of 105 studies reported an average no-show rate of about 23%, ranging from 13.2% (Oceania) to 43.0% (Africa), with long lead time and a prior no-show history as the main determinants [6]. These figures come from the review's abstract as reproduced on a secondary site and should be checked against the original before citing the numbers in the final report.

Open-access scheduling is one tested response. In a 2024 systematic review, 16 of 23,403 screened studies met the criteria; 10 (62.5%) reported a significant drop in noshows and 4 (25%) did not. The reviewers attribute success to needs assessment, system design based on actual patient and provider needs, stakeholder cooperation and consistent training [7]. A separate review of 50 articles covers statistical prediction of no-shows to improve appointment allocation [8].

Open Dental's manual shows a practical pattern: an Unscheduled List holds appointments that need follow-up, such as broken, no-show and cancelled ones, and a Confirmations tool tracks upcoming appointments and the action taken [17].

Design implication: model an appointment status (scheduled, confirmed, completed, cancelled, no-show), record patient, provider, date, time and duration, reject overlapping bookings per provider, and keep a follow-up list. Overbooking policy, open-access rules and no-show prediction depend on the clinic's policy [7] and are left out of the prototype unless the clinic asks for them.

Theme C — Treatment Plans and Clinical Documentation (RQ1, RQ2)

Treatment planning is treated as an essential element of both dental education and dental service [5]. In the axiUm EHR studied at two dental schools, a student obtains instructor approval and patient consent before beginning treatment [5]. Open Dental separates the stages explicitly: procedures are charted with

a treatment-planned (TP) status, treatment is prioritised in a Treatment Plan module, and payment-plan credits are only applied once a procedure is set complete [17].

Design implication: store a proposed treatment plan and a performed procedure as different states with different timestamps and users. The software records information; the dentist makes the clinical decision, and the prototype does not generate diagnoses.

Theme D — Security, Privacy and Access (RQ4)

A 2018 review of 20 studies from 10 countries found that role-based access control (RBAC) dominates health information systems, with 17 studies proposing adaptations. The main limitations were emergency access, grant delegation and cross-domain access control [9]. NIST SP 800-53 Rev. 5 defines least privilege as allowing only the accesses needed to accomplish assigned tasks (control AC-6), and has a separate audit and accountability family [10].

Production systems implement these ideas concretely. Open Dental assigns permissions to user groups, creates an audit-trail entry on every login and logoff, locks an account for five minutes after more than five consecutive failed logins, supports automatic logoff, and offers a lock date that prevents alteration of historical data [17]. OpenEMR configures access by role under Administration → ACL, with separate rights to view, add and modify patient demographics and notes, and extra access for sensitive encounters [18].

Locally, Egypt's Personal Data Protection Law No. 151 of 2020 treats data revealing physical, mental or genetic health as sensitive personal data, and treats children's data as sensitive in all cases [11]. Summaries of the law state that processing requires the data subject's explicit consent unless the law authorises otherwise, that data collected online must be encrypted in transit, and that large-scale or sensitive processing requires a Data Protection Officer [11][12]. These

summaries are secondary: check the official text before deployment. HIPAA is U.S.-specific and is used here only as comparative guidance.

Design implication: authenticate every user, enforce role-based permissions with least privilege, log access and changes, lock accounts after repeated failures, serve the application over HTTPS, and use synthetic data only. Because a dental clinic treats children, treat every patient record as sensitive.

Theme E — Usability and Web Architecture (RQ5)

Usability is a documented weakness of dental records. In a mixed-methods study of a widely used dental EHR interface, only 22–41% of 32 test users completed a simple task of entering one diagnosis, no user completed a more complex task, and the team characterised 24 high-level usability problems [13]. Usability tests of four dental computer-based patient record systems with novice users identified 286 usability problems [14]. A System Usability Scale (SUS) evaluation of the Salud EDR in Riyadh points to clearer icons, terminology and navigation paths [15], and a 2025 SUS study of a teaching-clinic EHR concludes that the record is useful but needs targeted training and interface changes [16].

None of the sources reviewed compares ASP.NET Core, Entity Framework Core and SQL Server with other stacks. The stack is therefore justified by team skills, maintainability and the built-in authentication and authorisation support, not by the literature. Open Dental's single main window with seven modules (Appointments, Family, Account, Treatment Plan, Chart, Imaging, Manage) is a useful navigation reference [17].

Design implication: keep workflows short, validate inputs, use plain dental terminology, and run a small usability test with a SUS questionnaire on the finished prototype.

2.6 Existing-System Comparison

Two real systems were compared using only their official documentation: Open Dental, a dental practice-management system, and OpenEMR, a general medical EHR with practice management. A cell states what the documentation pages we opened show.
"Not found" means the pages reviewed did not show it, not that the system lacks it. Neither system was installed or tested hands-on.

Criterion	Open Dental	OpenEMR	Proposed DCMS
Patient profile management	Family Module holds patient information and insurance plans [17]	Access control covers viewing, adding and modifying patient demographics [18]	Planned

Appointment scheduling	Appointments Module, recall, Unscheduled List and Confirmations [17]	Access control includes scheduling of appointments; wiki pages for calendar settings and a Patient Flow Board [18]	Planned
Dental treatment plans	Treatment Plan Module; procedures charted with TP status; Chart Module [17]	Not found: general medical EHR, no dental treatmentplan module in pages reviewed	Planned, simplified (proposed vs performed)
Role-based permissions	User groups with permissions, optional clinic restrictions [17]	Role-based ACL under Administration → ACL [18]	Planned
Audit logging	Audit trail on login, logoff and permissioncontrolled actions [17]	Wiki has an "Auditing in OpenEMR" page and auditreport pages [18]	Planned
Reports	Standard Reports under a Reports permission; Appointments Report [17]	Audit reports listed; other report types not verified	Planned, basic (daily appointments)
Patient portal	Offered through eServices [17]	Patient Portal has its own wiki page and ACL entry [18]	Out of scope unless approved

Two observations follow. First, Open Dental, a mature dental product, organises its work around seven modules and separates treatment planning, charting and scheduling. Second, both systems put role-based permissions and audit logging in the core, which supports treating them as baseline requirements and not optional extras.

scheduling. Second, both systems put role-based permissions and audit logging in the core, which supports treating them as baseline requirements and not optional extras.

2.7 Research Gap and Project Justification

The comparison does not show a functional gap: Open Dental already covers patient records, scheduling, treatment planning, permissions and reports [17]. The justification for this project is therefore about scope and learning, and does not claim that existing systems lack these features.

The review supports a deliberately limited prototype for three reasons. First, mature systems are broad: Open Dental spans seven modules, including insurance claims and imaging [17], which is far beyond a student team's capacity. Second, the evidence stresses a small set of high-value capabilities: complete baseline records [3], appointment status tracking [6][7], a clear split between planned and performed treatment [5][17], and role-based access with auditing [9][10]. Third, usability studies show that dental interfaces fail on terminology and navigation [13][14][15], so a narrow system tailored to one defined clinic workflow can be tested and refined more directly.

The project is therefore an educational prototype tailored to a defined dental-clinic workflow, with documented requirements, role-based access and synthetic data. Whether the target clinic has needs not covered by these systems must be confirmed with the clinic in Phase 3 requirements gathering.

2.8 Literature Review Summary

The literature supports a small, well-scoped system built on five findings, each mapped below to a candidate requirement for Phase 3. Requirement IDs are suggestions for the team to adopt or rename.

Theme	Finding	Source	Candidate Phase 3 requirement
Records	A baseline dental record needs demographics, medical and dental history, examination, treatment plan and progress notes	[3]	FR-01: The system shall store these elements per patient
Records	Surfacing a condition alert on opening a record helps staff notice it	[4]	FR-02: The system shall show a medical-alert banner on the patient record
Scheduling	No-shows are common, and appointment tracking is the basic response	[6][7]	FR-03: The system shall keep an appointment status and reject overlapping bookings per provider
Scheduling	Cancelled, broken and noshow visits need follow-up	[17]	FR-04: The system shall list appointments needing follow-up
Treatment	Proposed treatment and performed procedures are separate stages	[5][17]	FR-05: The system shall record planned and completed procedures separately, with user and time
Security	RBAC with least privilege is the standard model	[9][10]	NFR-01: The system shall enforce role-based permissions (admin, dentist, receptionist)

Security	Audit trails and login lockout are core features of mature systems	[17][18]	NFR-02: The system shall log logins and changes to clinical data and lock accounts after repeated failures
Privacy	Health data is sensitive under Egyptian law, and children's data always is	[11][12]	NFR-03: HTTPS only; synthetic data only during development and demonstration
Usability	Dental interfaces fail on terminology and navigation	[13][14][15][16]	NFR-04: Validate inputs, use plain dental terms and run a SUS test on the prototype

The main limits of this review are that it relied on abstracts and official pages and not on every full text, that no system was tested hands-on, and that the Egyptian legal sources are secondary summaries. The first two points are the team's next reading task; the third needs the official text of Law 151 of 2020.

2.9 References

Each entry lists the title, venue and year as shown on the source page, plus a link. Author names that the page snippets did not show are marked **[add authors]**; fill them in from the source, then format every entry in the citation style your course requires.

#	Reference	Link
1	Electronic Health Records in Dentistry: Relevance, Challenges and Policy Directions (review drawing on FDI World Dental Federation statements). [add authors, journal, year]	PMC12552933
2	Ballester B, Bukiet F, Dufour J-C. Current state of dental informatics in the field of health information systems: a scoping review. BMC Oral Health. 2022;22:131. doi:10.1186/s12903-022-02163-9	PMC9020044
3	A Qualitative Investigation of the Content of Dental Paper-based and Computer-based Patient Record Formats. [add authors, journal, year]	PMC2244908

4	A Digital Clinical Records Versus Paper Records in Dental Practice: A Comparative Study (Scarano et al.). BioMed Research International, 2026. [verify authors and citation details]	PMC12745653
5	Treatment Planning in Dentistry Using an Electronic Health Record: Implications for Undergraduate Education. [add authors, journal, year]	PMC3539210
6	Dantas LF, Fleck JL, Cyrino Oliveira FL, Hamacher S. No-shows in appointment scheduling – a systematic literature review. Health Policy. 2018;122(4):412–421. doi:10.1016/j.healthpol.2018.02.002	DOI
7	Mazaheri Habibi MR, Mohammad Abadi F, Tabesh H, Vakili-arki H, Abu-Hanna A, Ghaddaripouri K, Eslami S. Evaluation of no-show rate in outpatient clinics with open access scheduling system: a systematic review. Health Science Reports. 2024;7(7). doi:10.1002/hsr2.2160	PMC11231932

#	Reference	Link
8	Patient No-Show Prediction: A Systematic Literature Review. Entropy. 2020;22(6):675. doi:10.3390/e22060675 [add authors]	PMC7517206
9	Health Information System Role-Based Access Control Current Security Trends and Challenges. Journal of Healthcare Engineering. 2018. doi:10.1155/2018/6510249 [add authors]	PMC5836325
10	NIST. Security and Privacy Controls for Information Systems and Organizations, SP 800-53 Rev. 5 (AC-6 Least Privilege; Audit and Accountability family).	NIST CSRC
11	Arab Republic of Egypt. Law No. 151 of 2020 promulgating the Personal Data Protection Law (English text hosted by Andersen in Egypt). Secondary copy: verify against the official gazette.	PDF

12	Ibrachy & Dermarkar. Personal Data Protection Law No. 151 for year 2020, Arab Republic of Egypt (law-firm summary).	Article
13	Detection and characterization of usability problems in structured data entry interfaces in dentistry. [add authors, journal, year]	ASU record
14	Usability tests of four dental computer-based patient record systems. [add title as shown on page, authors, journal, year]	PMC2614265
15	Alnafrani S, Alsaeed S, Alsalwah NH, Alshammari A, Awawdeh M. Usability Assessment of Salud Electronic Dental Record System. [add journal, year]	Record
16	Souza JV, et al. Usability assessment of the electronic health record used in the school clinic of a higher education institution. Braz J Dent Educ. 2025;25:2458. doi:10.30979/BrazJDentEduc.v25.2458	Journal
17	Open Dental Software Inc. Open Dental manual: Modules, Using Open Dental, Security, User Group, Users, Permissions, Version 24.2 notes.	opendental.com
18	OpenEMR Project Wiki. Access Controls Listing; User Guide index; page index.	open-emr.org